Cosmic Visibility and Sampling Framework

An ASTRA research draft for operator-aware cosmology

Jacko T.

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> **Research status.** This is a repository-visible, unpromoted ASTRA supplemental research draft. It is not peer reviewed, is not a new SPPT equation, and does not report a dark-matter, graviton, axion, or planetary detection. It combines two supplied reports into a falsifiable measurement framework. The words *established*, *reported*, *inferred*, *proposed*, and *unknown* retain their ordinary evidence meanings throughout.

Abstract

Two recent case studies expose opposite failures of naive observation. In one, cosmic filaments are proposed as a distributed transducer: a hidden decay may produce a carrier that is converted by magnetized plasma into photons. In the other, a Martian meteorite fills an apparent age gap, showing that a missing class in the terrestrial collection need not be a missing class in the planet's history. The physical mechanisms are unrelated. The inference problem is the same.

The observable is produced by a chain rather than by a source in isolation:

$$\text{hidden state} \rightarrow \text{production} \rightarrow \text{transduction} \rightarrow \text{propagation} \rightarrow \text{archive or sampling} \rightarrow \text{detector} \rightarrow \text{certificate}.$$

This draft names the chain an **operator-aware visibility framework**. It extends the ASTRA bookkeeping idea (source, boundary, archive, observation, and bounded certificate) without pretending that the analogy is a physical unification. A source is inferable only relative to an identified visibility operator. A null result is strong only when the expected visibility is quantified. A transformed record can be useful rather than lost when the transformation is modelled and tested.

The framework contributes five practical objects:

1. a typed visibility kernel;
2. a source-versus-visibility identifiability ledger;
3. a multi-messenger calibration protocol for cosmic filaments;
4. a sampling-bias protocol for planetary archives; and
5. a promotion ladder that converts analogy into an executable, held-out test.

All cosmological applications below are conditional proposals. They are not claims that dark matter decays, that gravitons have been detected, that a particular Martian rock defines a fourth mantle reservoir, or that the universe is a computer, ether, or single machine in a literal sense.

1. What changed, and what did not

1.1 The methodological advance

The useful change is a change in the unit of analysis. A conventional question asks, “Is the source present?” The framework asks:

- What state or history would produce the residue?
- Which physical carrier transports that residue?
- What medium converts or attenuates it?
- Which archive or sampling process selects the surviving record?
- What detector and nuisance model map the record to a certificate?
- Which combinations of source and visibility parameters are observationally equivalent?

This is **operator-aware reconstruction**. It does not add an unknown field to SPPT and it does not turn a metaphor into a law. It is a reporting and design layer that can be attached to a domain when the relevant operators have units, priors, calibration data, and falsification tests.

1.2 Explicit non-claims

The following are deliberately outside the admitted scope:

- a detection of dark matter or a measurement of its lifetime;
- a detection of a single graviton or graviton–photon conversion event;
- a detection of an axion or axion–photon conversion event;
- proof that cosmic magnetic fields have any one origin or universal profile;
- proof that NWA 13441 is a unique, pristine, or fourth Martian mantle reservoir;
- proof of a shared physical mechanism between cosmic conversion and meteorite delivery;
- a replacement for the standard cosmological model, QED, general relativity, isotope geochemistry, planetary petrology, or their experimental records; and
- a revision of the immutable SPPT/ASTRA v1.0.6 core or its release identity.

1.3 Relationship to SPPT and ASTRA

SPPT remains the phase-reservoir graph and its conservation, transport, boundary, topology, intervention, and held-out prediction contracts. ASTRA is the analysis layer that records hidden states, observation boundaries, provenance, and evidence status. This draft is a namespaced successor method:

$$\text{SPPT state model} \;+\; \text{ASTRA evidence ledger} \;+\; \text{visibility and sampling operators.}$$

The plus signs denote typed interfaces, not an assertion that the domains share a physical substance. No statement in this document is promoted into the v1.0.6 claim matrix merely by appearing here.

2. Two case studies, one inference boundary

2.1 Cosmic filaments as conditional transducers

The supplied dark-sector report follows a proposed chain in which an unstable particle χ decays to two gravitons, and a graviton can convert to a photon while traversing a magnetic field. In schematic form,

$$\chi \to h + h \to h \leftrightarrow \gamma \text{ in } B \to \gamma\text{-ray propagation} \to \text{all-sky background.}$$

The important result is a conditional null constraint: under a specified abundance, branching fraction, filament field distribution, coherence length, propagation model, and background decomposition, an excess gamma-ray component would not be allowed above a calculated level. This is not a detection of any link in the chain.

For a short coherent path, the proposed conversion probability has the schematic scaling

$$P^{(1)}_{h \to \gamma} \propto G B_{\perp}^{2}\ \ell_{\text{coh}}^{2}\ , \qquad N_{\text{fil}} \propto f_{\text{vol}}\ \frac{L}{\ell_{\text{fil}}},$$

so a cumulative sensitivity can scale approximately as

$$P^{\text{total}}_{h \to \gamma} \propto B_{\perp}^{2}\ \ell_{\text{coh}}^{2}\ f_{\text{vol}}\, L/\, \ell_{\text{seg}}\ ,$$

where the shorter segment length ℓ_{seg} is explicit; the often-used linear ℓ_{coh} scaling follows only when ℓ_{seg} is set equal to the coherence length. These proportionalities are design intuition, not a substitute for the coupled wave calculation. They reveal why the filament is part of the instrument. A stronger cluster field does not automatically dominate if clusters occupy less volume or provide a shorter coherent path. The expected signal is a compound function of source abundance, lifetime, branching fraction, magnetic field, geometry, plasma, redshift, and detector response.

2.2 NWA 13441 as a selective archive

The supplied Mars report describes NWA 13441 as an olivine-phyric shergottite with a reported Sm–Nd errorchron age near 1273 ± 21 Ma. The age lies between large clusters of younger and much older shergottites. The safe inference is that the terrestrial collection had not previously sampled that part of the Martian magmatic record. It is not safe to infer that Mars lacked magmatism in the interval.

The delivery chain is long and selective:

$$\text{mantle source} \to \text{melt and crystallization} \to \text{impact excavation} \to \text{ejection} \to \text{interplanetary transfer} \to \text{terrestrial survival} -$$

Each arrow can remove, transform, or preferentially retain material. The reported isotopic composition is compatible with more than one generator, including a long-lived near-chondritic reservoir and mixing among known reservoirs. The available record does not yet identify which generator is correct. A large regression scatter or excluded alteration-affected fractions should be retained in the uncertainty ledger, not hidden by the headline.

2.3 Why the cases should not be physically conflated

Cosmic conversion and meteorite delivery are not evidence of one universal force. One is a proposed wave conversion in a magnetized medium; the other is a geological and impact-selection history. Their commonality is structural:

Case-role comparison and inference risks

Case	Hidden object	Intervening operator	Terminal record	Main risk
Filament search	decay or hidden-sector residue	magnetic conversion, propagation, exposure	gamma/radio map	source–field degeneracy
Martian archive	mantle history	melting, launch, transfer, recovery	small rock collection	archive and sampling bias

This is exactly the level at which ASTRA can be useful: typed interfaces and test plans, not unsupported ontology.

3. The visibility kernel

3.1 Definition

Let H denote a source history, q a latent coordinate (space, time, frequency, energy, composition, or geometry), and m a messenger or assay. Define the visibility kernel as an operator composition first:

$$\mathscr{V}_m = \mathscr{O}_m \circ \mathscr{S}_m \circ \mathscr{P}_m \circ \mathscr{T}_m, \qquad \mathbf{y}_m = \mathscr{V}_m[\mathscr{R}_H] + \mathbf{f}_m + \boldsymbol{\varepsilon}_m.$$

When the operators reduce to independent scalar efficiencies at a declared resolution, a useful approximation is

$$\eta_m(q;\psi) = \eta_{\text{prod}}(q)\,\eta_{\text{trans}}(q;\psi_T)\,\eta_{\text{prop}}(q;\psi_P)\,\eta_{\text{arch}}(q;\psi_A)\,\eta_{\text{samp}}(q;\psi_S)\,\eta_{\text{obs}}(q;\psi_O),$$

where ψ are nuisance and calibration parameters. This product is not an independence assumption about the physical processes: it is a conditional factorization that must be justified by the model and can fail for coupled, history-dependent, or selection-dependent operators. The factors may be probabilities, efficiencies, transfer operators, or integral kernels; they need not all be dimensionless. A unit-bearing forward model should state the measure and normalization explicitly rather than silently multiplying unlike quantities.

The expected observable yield under H is

$$\Lambda_{H,m} = \int r_H(q)\,\eta_m(q;\psi)\,dq,$$

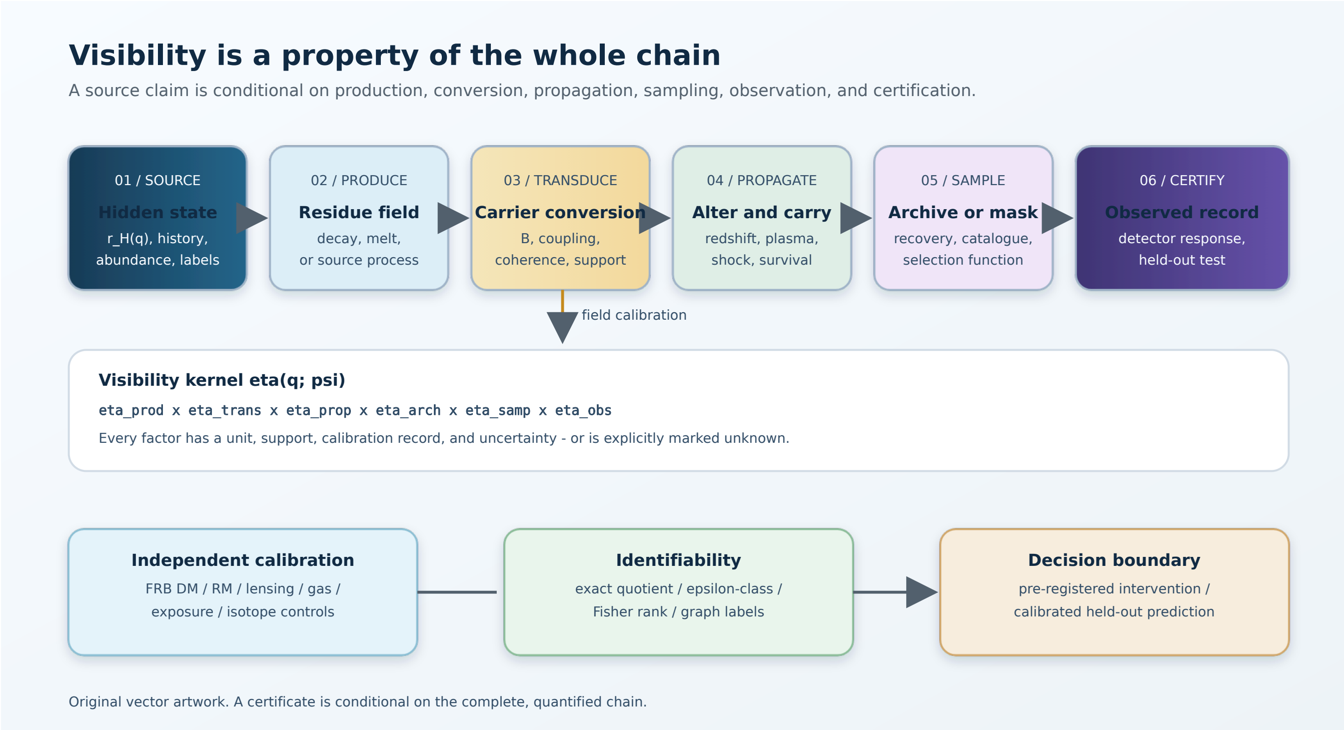
where r_H is the source or residue field in compatible units. In a discrete model, write $\boldsymbol{\lambda}_H = \mathbf{K}_m(\psi)\mathbf{r}_H$ and record the matrix, units, resolution, and boundary conditions.

3.2 Forward observation model

For a map, spectrum, or assay vector $\mathbf{y}_m$,

$$\mathbf{y}_m = \mathscr{O}_m[\mathscr{P}_m(\mathscr{T}_m[\mathscr{R}_H])] + \mathbf{f}_m(\nu_m) + \boldsymbol{\varepsilon}_m.$$

Here $\mathscr{T}$ is transduction, $\mathscr{P}$ is propagation and alteration, $\mathscr{O}$ is detector or assay response, $\mathbf{f}$ is a nuisance or foreground model, and $\boldsymbol{\varepsilon}$ is measurement error. For a sample archive, $\mathscr{O}$ includes laboratory preparation and measurement, while $\mathscr{S}$ governs which fragments enter the archive at all.



The visibility kernel separates source production, physical conversion, propagation, archival or sampling selection, and the terminal observation. The highlighted certificate is conditional on the complete chain.

3.3 Absence and the zero-count limit

For a simple Poisson count, $N \sim \text{Poisson}(\Lambda_H)$ and

$$P(N=0 \mid H) = e^{-\Lambda_H}, \qquad V_0(H) = -\log_{10} P(N=0 \mid H) = \frac{\Lambda_H}{\ln 10}.$$

The formula is useful because it forces visibility into the null claim. It is not universally applicable: clustered counts, correlated backgrounds, selection on detection, and model uncertainty require a likelihood or point process appropriate to the data. If η is poorly constrained, the null usually identifies a product or integral of source and visibility parameters, not the source parameter alone.

3.4 Visibility is channel-relative

“Dark” is not an absolute label. A state can be dark to one detector and visible to another after carrier conversion:

$$\text{hidden state} \to \text{charged residue} \to \text{synchrotron}, \qquad \text{hidden state} \to h \to \gamma, \qquad \text{hidden state} \to a \to \gamma.$$

These chains are distinct hypotheses. The framework does not combine them into a common particle. It requires each to specify its own coupling, normalization, propagation, and rejection tests.

4. Identifiability and the source–visibility quotient

4.1 Exact and practical equivalence

Let θ describe source history and ψ describe the visibility field. Two parameter pairs are **exactly observationally equivalent** on an experiment $\mathscr{E}$ if

$$\mathscr{F}_{\mathscr{E}}(\theta_1, \psi_1) = \mathscr{F}_{\mathscr{E}}(\theta_2, \psi_2)$$

for every observable in the declared data space, including uncertainty model and support. They are ϵ**-equivalent** on a finite design if the distance between predicted distributions is below a predeclared tolerance:

$$d(P_{\theta_1, \psi_1}, P_{\theta_2, \psi_2}) \leq \epsilon.$$

The metric d may be a whitened prediction norm, likelihood-ratio distance, energy distance, or a domain-specific discrepancy. The choice and scale are part of the protocol, not a post-hoc convenience.

The **visibility quotient** is the set of equivalence classes under this relation. A result that identifies only a quotient class must not be worded as identifying a unique source.

4.2 Local information and rank

For a differentiable mean response $\boldsymbol{\mu}(\theta, \psi)$ and covariance $\boldsymbol{\Sigma}$, the local Fisher information is

$$\mathbf{I}(\vartheta) = \mathbf{J}(\vartheta)^{\top} \boldsymbol{\Sigma}^{-1} \mathbf{J}(\vartheta), \qquad \mathbf{J} = \frac{\partial \boldsymbol{\mu}}{\partial \vartheta^{\top}}, \quad \vartheta = (\theta, \psi).$$

Rank deficiency or a tiny singular value indicates local non-identifiability under the chosen design. It does not prove global equivalence, and full local rank does not protect against an unmodelled channel. Report singular values, scaling, parameter units, and the perturbation size used to estimate $\mathbf{J}$.

4.3 Graph labels and topology quotienting

When the hidden system is a graph, relabeling nodes can produce exactly the same observables. A graph claim should therefore define the admitted label group G and compare graph hypotheses modulo the action of G:

$$[G] = \{g \cdot G : g \in \mathscr{G}_{\text{labels}}\}.$$

For weighted or typed graphs, the quotient must preserve edge direction, channel type, boundary labels, units, and intervention ports. A graph isomorphism test is not a proof that two physical systems are equivalent; it only removes a bookkeeping redundancy before response comparison.

4.4 Interventions as designed separation

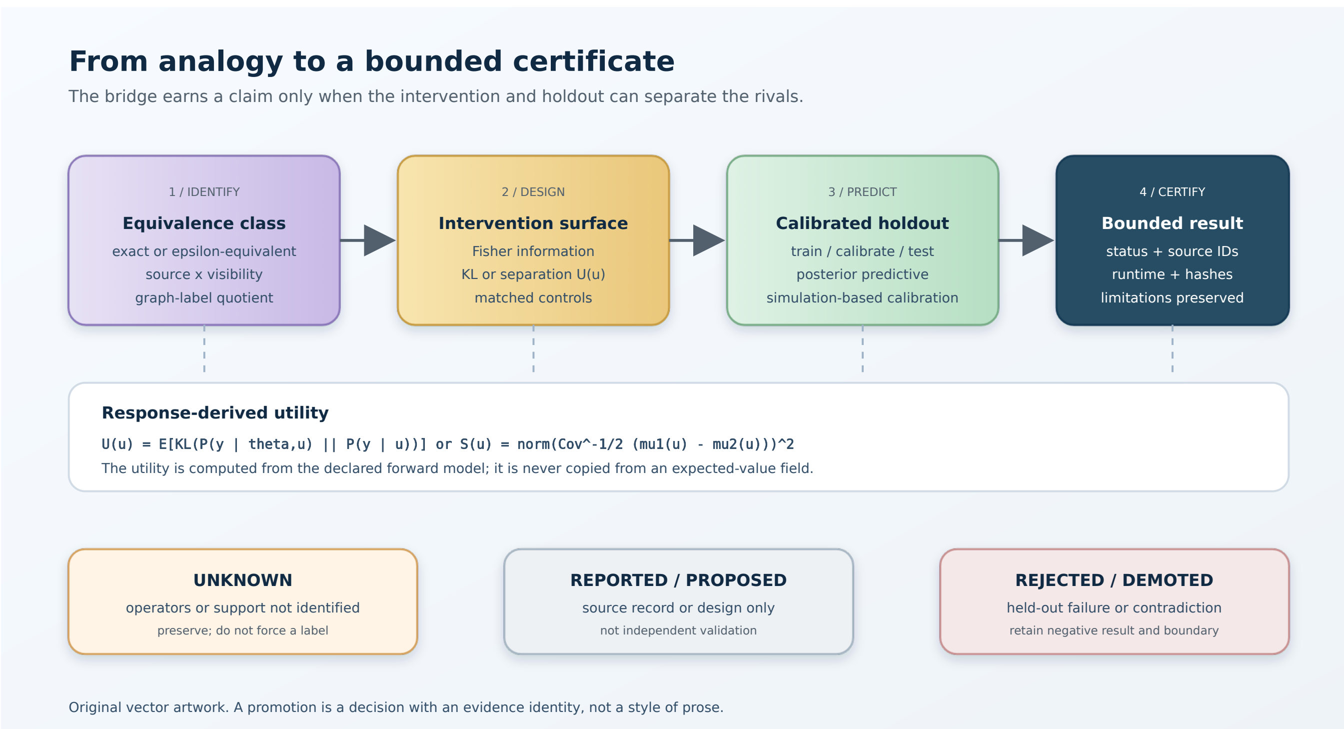
An intervention u changes forcing, exposure, mode, sampling, or a calibrated medium parameter. Choose u by the expected separation of rival predictions, not by a supplied utility number. One practical score is

$$U(u) = \mathbb{E}_{\vartheta \sim \pi}[\ \mathrm{KL}\ (P(\mathbf{y} \mid \vartheta, u)\ \|\ P(\mathbf{y} \mid u))],$$

or, for two hypotheses, a noise-whitened separation

$$S(u) = \ \| \boldsymbol{\Sigma}_u^{-1/2} (\boldsymbol{\mu}_1(u) - \boldsymbol{\mu}_2(u)) \|_2^2 .$$

The response surface must be generated from the forward model over u and the admissible parameter set. A hand-entered expected gain is evidence of a design choice, not evidence that the design works.



An intervention plan should move from an observational equivalence class to a response surface, then to a calibrated, held-out certificate.

5. The cosmic web as a calibrated instrument

5.1 Typed roles

For a filament ensemble, record the roles separately:

Role ledger for a calibrated filament instrument

Role	Example quantity	Failure if omitted
Source	ρ_{DM}, decay rate, branching fraction	false abundance or lifetime claim
Transducer	$B_{\perp}$, coupling, coherence length	source–field degeneracy
Propagator	redshift, plasma, pair cascades, attenuation	wrong spectrum or morphology
Amplifier or emitter	synchrotron particles, resonance, cascade	wrong energy budget
Sampler	filament catalogue, mask, sightline selection	catalogue bias
Archive	gamma, radio, FRB, lensing, RM maps	missing or correlated evidence
Detector	exposure, PSF, energy response, calibration	invalid significance
Certificate	predeclared test statistic and holdout	post-hoc story

The same physical filament can carry more than one role, but each role gets a separate field and uncertainty record.

5.2 Calibration channels

No filament-mediated search should fix one representative magnetic field and then treat the result as source-only. The minimum calibration vector should include, where available,

$$\Sigma_F = (\rho_{\mathrm{DM}}, n_e, T_e, B_{\parallel}, B_{\perp}, \ell_{\mathrm{coh}}, z, G_F),$$

with geometry and orientation G_F. Candidate tracers have complementary operators:

$$\mathrm{DM}_{\mathrm{FRB}} \propto \int n_e \, dl, \qquad \mathrm{RM} \propto \int n_e B_{\parallel} \, dl,$$

$$I_{\mathrm{sync}} \propto \int n_{e^{\pm}} B_{\perp}^{(p+1)/2} \, dl, \qquad \kappa_{\mathrm{lens}} \sim \int \rho_{\mathrm{total}} W_{\kappa} \, dl.$$

These observables do not measure the same thing. Their value is in jointly constraining the nuisance field and exposing incompatible assumptions.

5.3 Filament Conversion Tomography protocol

The following is a proposed experiment design, not a result.

Inputs

1. A filament or web catalogue with a frozen version, selection function, redshift range, and geometry uncertainty.
2. Gamma-ray and radio maps with exposure, point-spread, energy/frequency response, foreground masks, and time range.
3. FRB dispersion and Faraday-rotation records with source selection and calibration metadata.
4. Weak-lensing or CMB-lensing maps and a constrained simulation ensemble.
5. A null-control catalogue with matched converter-proxy strata and void-dominated baryon comparison sightlines. A void is not a low-field or zero-dark-matter control unless independent magnetic and matter proxies establish that interpretation.

Pre-registration

Before opening the final residual map, freeze:

- the photon or radio spectrum and its mass or coupling dependence;
- the redshift and angular weighting;
- the dependence on $B_{\perp}^2$, any separately modelled source-column proxy, and coherence length;
- the treatment of unknown or partially observed fields;
- primary and secondary cross-correlation statistics;
- the holdout split and nuisance prior; and
- the threshold for rejecting, demoting, or retaining the candidate.

Primary test

Estimate a joint model for gamma/radio residuals and independent filament tracers. A candidate conversion signal should produce a predeclared spatial or tomographic relationship with filament probability and magnetic proxies. It should not be equally strong in matched low-converter controls after exposure and foreground adjustment. Void sightlines are a baryon and sampling comparison; they become a magnetic-control stratum only when an independent RM-squared or equivalent proxy supports that interpretation.

Rejection conditions

Demote the candidate if any of the following occurs in a calibrated, held-out test:

- the residual does not track filament probability when sensitivity is adequate;
- the residual does not scale with independent magnetic-field proxies;
- the energy spectrum or redshift dependence disagrees with the forward model;
- an ordinary source population explains the same morphology and spectrum;
- matched low-converter controls show the same effect, or void controls show the same effect after their baryon, magnetic, and selection differences are explicitly modelled;
- the required field conflicts with FRB, RM, radio, cascade, or laboratory constraints; or
- no single parameter set closes across the abundance, structure, and coupling archives.

5.4 Interpreting a null

A Fermi-LAT-like null can be a meaningful constraint only after integrating the field uncertainty. In a simplified model,

$$\Phi_\gamma \propto \frac{b_{\text{hidden}}}{m_\chi \tau_\chi} \int dz\, \rho_\chi(z) B_\perp^2(z)\, \ell_{\text{coh}}(z) K(E, z).$$

The null therefore constrains the compound integral. If b_{hidden} or B is free over orders of magnitude, a lifetime bound cannot be described as model-independent. The release of a null certificate should include posterior or profile sensitivity to each visibility component.

6. The planetary archive as a sampling experiment

6.1 Archive-entry operator

For an underlying population of Martian source rocks with density $r_M(q)$, let $\mathscr{A}$ denote the archive-entry, preservation, recovery, and recognition operator. Under a declared factorization, let $a(q)$ be the probability of entering the terrestrial archive and $s(q)$ the probability of being recovered and recognized. The expected observed sample is

$$r_{\text{observed}}(q) = \mathscr{A}[r_M](q), \qquad \mathscr{A}[r_M](q) = r_M(q)\,a(q)\,s(q) \quad \text{only under that factorization.}$$

A gap in r_{observed} is evidence against r_M only when $a(q)s(q)$ is bounded away from zero in the relevant region. For shergottites, impact history, launch geometry, strength, transit, weathering, human collection, and recognition are all selection factors.

6.2 Competing generators for NWA 13441

The present draft keeps at least two generators visible:

1. a long-lived, weakly mixed near-chondritic mantle reservoir; or
2. mixing of enriched and depleted melts that reproduces a near-chondritic isotope value.

The Sm–Nd result is a bounded chronology and compositional constraint, not a unique source certificate. The following measurements can separate the models:

- Rb–Sr and Lu–Hf isotope systems;
- mineral-scale isotope maps and heterogeneity tests;
- cosmic-ray exposure and noble-gas measurements;
- diffusion and shock-reset modelling;
- comparison with chemically paired and unpaired shergottites;
- impact-launch and ejection-site models; and
- a systematic search for additional rocks in the age interval.

The high-information test is not “find another unusual rock.” It is to obtain independent isotope trajectories and source context that produce different predictions under isolation and mixing.

6.3 Sampling controls

An archive-aware protocol should report:

- the candidate source population and its unobserved support;
- the transport and preservation mechanisms;
- the recovery and recognition process;

- known and suspected collection biases;
- a missing-not-at-random sensitivity analysis; and
- a proposed observation that would change the source-versus-sampling odds.

This is a direct analogue of the void control in a cosmic-web search: a comparison class tests whether the apparent signal or gap follows the proposed operator rather than only the terminal record.

7. Genesis and coupling are different archives

A dark-sector hypothesis needs two consistency ledgers.

7.1 Genesis archive

Production history can affect power spectra, isocurvature, free-streaming, halo abundance, cosmic-web topology, CMB distortions, and gravitational-wave backgrounds. A coupling model that fits a present gamma-ray channel but cannot produce the observed abundance or structure is incomplete.

7.2 Coupling archive

Present interactions can leave gamma rays, radio emission, conversion or absorption features, scattering, stellar heating or cooling, and other messenger-specific residues. A genesis model that fits abundance but predicts an excluded coupling is likewise incomplete.

The joint gate is:

$$\text{genesis consistency} \;+\; \text{present-coupling consistency} \;+\; \text{visibility calibration.}$$

No one archive substitutes for the others.

8. Field-level and population-level reconstruction

8.1 Resolved objects are not the only sample

Galaxy catalogues preferentially contain objects that are luminous, extended, high surface-brightness, and observable through the survey mask. Line-intensity mapping and unresolved-background analyses instead estimate aggregate fields. This can recover information from populations below individual detection thresholds, but it moves the burden to interloper, foreground, and transfer models.

An interloper can become a useful tracer when its redshift and line response are modelled jointly. If that operator is wrong, the same “extra information” can bias cosmological parameters. Renaming contamination as signal is not a calibration.

8.2 Phases, topology, and spin

The present universe stores information in more than two-point amplitudes. Candidate archives include Fourier phases, filament connectivity, void structure, persistent homology, halo histories, and spin alignments. These channels can help break degeneracies, but they also have selection functions and simulation-model dependence.

The safe statement is that non-Gaussian and topological summaries are promising additional observables. Their information gain must be measured against a frozen baseline on held-out simulations or observations.

8.3 Posterior checks are necessary but not sufficient

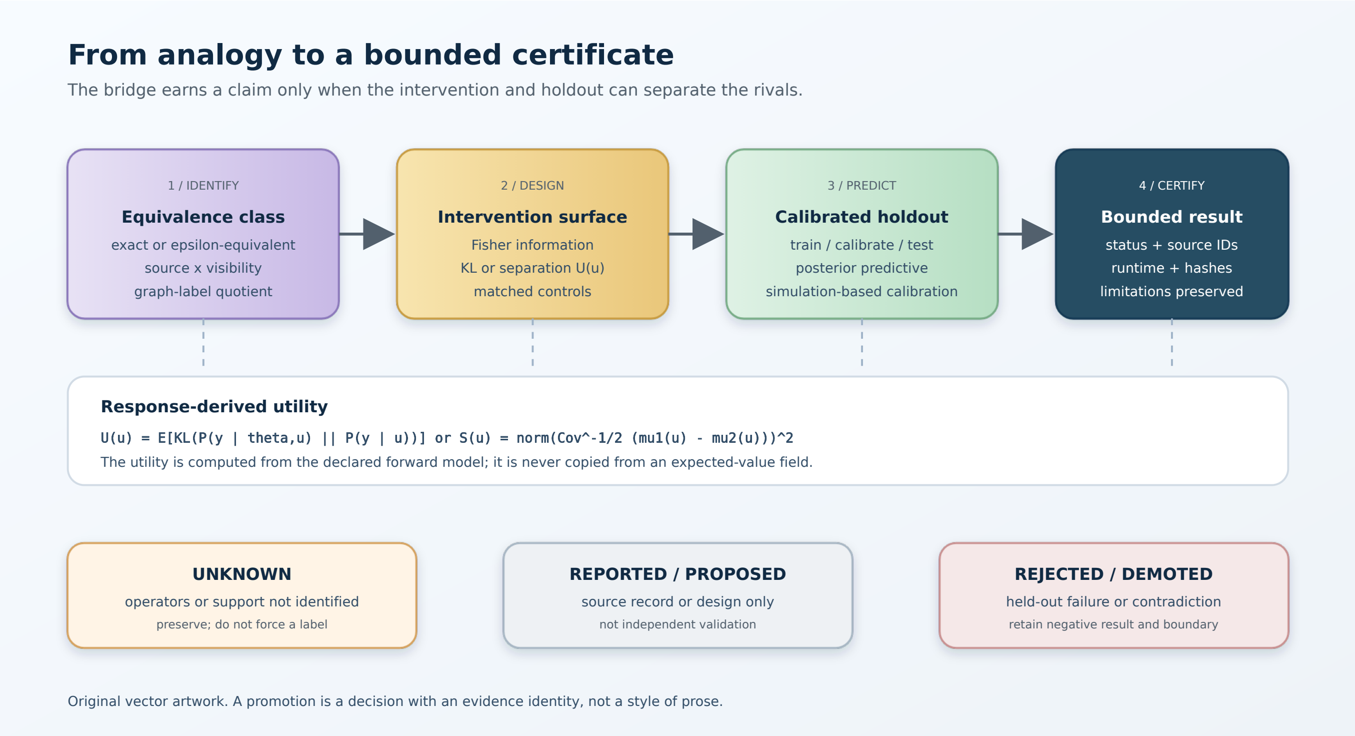
Simulation-based or field-level inference can be confidently wrong when the digital twin absorbs model error into nuisance fields. A robust protocol needs:

- seeded train, calibration, and held-out splits;
- posterior-predictive checks at fixed parameters;
- simulation-based calibration with known generators;
- out-of-model and omitted-channel controls;
- nuisance-shift and coverage diagnostics; and
- an explicit failure record when the diagnostic itself is underpowered.

Coverage of one interval is not proof that the forward model is complete.

9. The unified protocol

The proposed protocol is a typed sequence rather than a claim about one physical law.



The promotion ladder separates source claims, operator calibration, identifiability, interventions, and held-out certificates.

Step A: conservation and provenance contract

Declare the state, units, boundaries, source records, artifact hashes, retrieval dates, and rights. For an archive, record the sampling frame and missingness mechanism. For a cosmic map, record exposure, mask, calibration, and simulation version.

Step B: thermodynamic or budget ledger

Track energy, mass, charge, isotope inventory, photon counts, or the relevant conserved quantity through each operator. Where a thermodynamic interpretation is appropriate, record entropy production and exergy separately from analogy. No conversion efficiency may be counted twice.

Step C: identifiability class

Compute exact equivalence where possible; otherwise report practical classes under a named discrepancy metric. Quotient graph labels and nuisance symmetries. Report Fisher singular values, controllability/observability ranks where a state-space model applies, and the support of each data channel.

Step D: designed intervention

Use response surfaces and Fisher or separation metrics to choose forcing, frequency, sightline, isotope system, field proxy, or matched control. The intervention must be specified before the held-out result is inspected.

Step E: calibrated held-out prediction

Fit only on the training split. Calibrate nuisance and uncertainty on the calibration split. Score held-out data with a predeclared discrepancy, proper scoring rule, interval coverage, and posterior-predictive check. Preserve negative synthetic and real results.

Step F: promotion decision

Use evidence labels:

Evidence labels used by this draft

Label	Meaning in this draft
Established	directly supported by a primary record or reproducible measurement
Reported	accurately summarized from a named source, without independent replay
Inferred	methodological conclusion from multiple records or equations
Proposed	a testable design or model extension
Unknown	the available evidence cannot distinguish alternatives
Rejected	contradicted by a declared test or outside its domain

Promotion requires source-local support, machine-readable metadata, an exact runtime or archive identity, an independent replay where claimed, and a held-out result that beats the declared baseline without leaking evaluation data into design.

10. Concrete research program

The following work packages are ordered by information value and bounded by the framework's evidence rules.

WP1: filament field calibration

Build a versioned filament catalogue and infer B, n_e, coherence length, geometry, and uncertainty from FRB DM, RM or RM-squared, synchrotron, X-ray, lensing, and galaxy data. Hold out sky regions and redshift bins. Report the posterior predictive distribution of each proxy before fitting a hidden-sector channel.

WP2: converter-specific forward models

Implement separate models for charged decay products, graviton conversion, and axion conversion. Each must carry its coupling and units, include propagation and cascades, and expose the source–visibility quotient. The models must not share parameters merely because the diagrams look alike.

WP3: matched filament–void controls

Construct matched sightline pairs by exposure, redshift, angular mask, and ordinary source density. Vary filament probability and independent field proxy. Pre-register the expected direction and scale of the effect. A null in a low-visibility control cannot veto a high-visibility source.

WP4: Martian archive expansion

Measure independent isotope systems, exposure ages, noble gases, mineral-scale heterogeneity, and shock histories. Model launch and recovery selection. Score the isolated-reservoir and mixing generators on samples not used to tune the source models.

WP5: bridge benchmark

Create synthetic hidden states with known source and visibility operators, including deliberately omitted channels, anisotropic transport, nonstationary sampling, miscalibrated detectors, and hidden-state equivalence. Require the bridge to return the correct equivalence class or an explicit unknown—not a forced unique label.

WP6: adapter boundary to SPPT

Only after WP1–WP5 pass should a successor adapter map a domain-specific visibility contract onto SPPT's strict incidence and thermodynamic edge types. The adapter must preserve direction, units, conservation residuals, entropy inequality, and source-record identity. It must never retrofit a cosmological analogy into the immutable core.

11. Failure modes and safeguards

11.1 Source strength mistaken for visibility

Failure: A weak limit is reported as a weak source. **Safeguard:** publish sensitivity to every visibility factor and the posterior or profile of the compound parameter.

11.2 Archive gap mistaken for historical absence

Failure: An unsampled age or composition interval is treated as a physical gap. **Safeguard:** model archive-entry and recovery selection; seek independent samples and source context.

11.3 Background or interloper relabelled as signal

Failure: a flexible nuisance model absorbs or creates a residual. **Safeguard:** hold out regions, use independent channels, and score an ordinary source baseline with the same flexibility.

11.4 Analogy promoted to mechanism

Failure: “instrument,” “pressure,” “memory,” or “conversion” is treated as a physical law without units. **Safeguard:** require an equation, observable, calibration, and falsifier.

11.5 Post-hoc topology or source selection

Failure: the graph, filament sample, or isotope fractions are chosen after seeing the result. **Safeguard:** freeze the candidate set, selection function, and holdout before the final score.

11.6 Overconfident nulls

Failure: $N = 0$ is interpreted as decisive when η is unknown. **Safeguard:** report a visibility-conditional null and state the assumptions under which veto strength is nontrivial.

12. What would count as a genuine advance?

The framework itself is a methods proposal. A future scientific advance would need to clear a higher bar:

1. a primary record and exact source locator;
2. a unit-consistent forward model with independent implementation or replay;
3. a parameterized visibility field constrained by calibration channels;
4. a source-versus-visibility identifiability analysis;
5. a predeclared intervention or matched control;
6. a held-out prediction that beats an appropriate conventional baseline;
7. posterior-predictive, simulation-based calibration, and omitted-channel checks; and
8. a reproducible artifact whose source, runtime, hashes, and rights are public.

Passing these gates would justify a domain-specific result. It would not by itself prove a universal “cosmic instrument” ontology. The right reward for a successful bridge is a narrower, stronger claim.

13. Evidence ledger for this draft

Draft evidence ledger and next evidence

Statement	Status	Required next evidence
A hidden source can become observable after physical conversion	Inferred conditional modelling pattern; channel-specific realizations remain conditional	independent conversion calibration or domain-specific replay
Filament conversion can constrain a modelled dark-sector decay	Reported within declared assumptions	joint field calibration and held-out cross-correlation
Graviton or axion conversion has been observed in filaments	Unknown / unsupported	event-level or statistically discriminating evidence
Cosmic filaments contain baryons and matter	Established in broad terms	improved multi-messenger calibration
Filament magnetic fields can be used as a calibrated conversion proxy	Model-conditional and proxy-dependent; not a universal direct measurement	independent RM-squared, synchrotron, lensing, and plasma calibration
NWA 13441 fills a sampling gap in known shergottite ages	Reported from the supplied primary-study summary	independent samples and exposure/launch constraints
NWA 13441 uniquely identifies a pristine or fourth mantle reservoir	Unknown	multi-isotope and source-context separation of generators

Statement	Status	Required next evidence
Visibility and sampling should be modelled as operators	Inferred methodological contribution	benchmarked bridge with known hidden states
The universe and a meteorite collection share one physical mechanism	Rejected as an unsupported physical claim	none unless a new mechanism is proposed and tested

14. Reproducibility and rights boundary

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References and primary-record leads

The following links are leads supplied with the research reports. They are listed so a future audit can attach source IDs, exact locators, retrieval dates, and hashes. A link in this draft does not by itself establish entailment.

- Dark matter decay signals in cosmic filaments
- Observing dark matter decays to gravitons via graviton–photon conversion
- Probing the cosmic axion background via axion–photon conversion in filaments
- CMB spectral distortions from resonant conversions in atomic dark sectors
- Probing cosmic magnetism with rotation-measure-squared correlations
- Backlighting the cosmic web with fast radio bursts
- Implications for Martian mantle reservoirs from NWA 13441
- LPSC 2026 abstract on NWA 13441, preliminary context (conference values are not merged with the final paper)
- Baryons in the darkest sites of the universe
- Constraints on annihilating dark matter from gamma-ray cross-correlations
- Searching for Population III stars with line-intensity mapping
- Interlopers as signal in line-intensity mapping
- Primordial tidal-torque imprints
- Field-level inference of primordial non-Gaussianity
- Disentangling modified gravity and galaxy bias
- Cosmic-web topology and neutrino mass
- Confining dark sectors and cosmological perturbations
- Digital twins of the local universe
- Coverage tests for simulation-based inference

Closing principle

> **Infer the source and the visibility field together. Use independent messengers to calibrate the converter, matched low-visibility controls to test the sampler, and held-out predictions to decide what survives.**

Ad astra per aspera.